



IITM

MAD – II Project

----- Quiz Management App -----

Created By:

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Executive Summary :

The Ischool Quiz App is a robust, scalable Quiz Management System, designed to optimize the creation, deployment and evaluation of quizzes. Developed utilizing concepts from Modern App Development – 1 & 2, this app ensures seamless user experience for both administrators and participants, enhancing the efficiency of online assessment.

Technology Stack:

- **Backend:** Flask (Python) with Flask-SQLAlchemy for ORM-based database interaction.
- **Database:** SQLite for structured, lightweight data persistence.
- **Frontend Layer:** Vue.js for the dynamic user interface, styled with the Bootstrap framework.
- **Email Templates:** Jinja2 templating engine for generating dynamic email content on the backend.

Key Frameworks and Libraries:

- **Flask-CORS:** Enables secure communication between the frontend and the backend API.
- **Flask-Caching:** Improves application speed by caching frequently requested data in Redis.
- **Bcrypt:** Provides secure, one-way hashing and salting for user passwords.
- **JWT :** Generates stateless JSON Web Tokens for secure user authentication.
- **Redis:** High-performance in-memory data store for caching and for managing Celery's background task queue.
- **Celery:** Executes asynchronous background tasks, such as sending scheduled reports.
- **SMTP:** Protocol used with Google's App Password for dispatching automated emails.
- **Axios:** A promise-based HTTP client used on the frontend to interact with the Flask API.

Core Functionalities and Features:

Administrative Capabilities:

1. Full CRUD operations for level, subject, chapter, quiz, and question management.
2. Advanced analytics dashboard featuring real-time user engagement insights and quiz performance tracking.
3. User Management: Ability to activate, suspend, or delete users to maintain platform integrity.
4. Enhanced search and filtering, enabling administrators to retrieve specific records efficiently.

5. Auto email is sent to user for monthly active report and daily email are send for newly create quiz if it is relevant to user.

End-User Functionalities:

1. Secure Authentication & Authorization – Users register and authenticate using a robust credential system.
2. Personalized Dashboard displaying performance metrics, past quiz attempts.
3. Real-time Quiz Participation with an integrated timer and answer validation
4. Post-quiz analytics, including score breakdowns, correctness analysis, and improvement suggestions.
5. Comprehensive Query System – Users can search for quizzes by subject, chapter, date, and performance scores.
6. User can also delete their account.

Operational Workflow:

Administrative Process Flow

1. **Dashboard Access:** Admins are directly redirected to a secure control panel.
2. **Quiz Configuration:** Creation of level, subjects, chapters, and quizzes with custom parameters.
3. **Question Bank Management:** Definition of multiple-choice questions with associated correct responses and timers.
4. **User & Data Governance:** Performance monitoring, user access management, and report generation.

User Engagement Process

1. **Authentication & Access:** Users log in securely via the authentication system.
2. **Quiz Selection & Participation:** Users browse available quizzes and engage in real-time assessment.
3. **Answer Submission & Validation:** Responses are recorded, evaluated, and scored.
4. **Performance Review:** Users receive comprehensive results, including correctness breakdowns and improvement suggestions.

RUN CMD:

For Frontend: “npm run server”

For Backend: “app.py”

For Redis: “redis-server”

Database Schema

All Database schema is given shows in pictorial form.

